6 Algebra, equations and inequalities

Sheet 31 Thu 14 Jan. 2027 *Integer powers (positive and negative)*

1. Warm-up · Quick Calculate 2^{-3}, 10^{-2} and $(-3)^2$.
2. Core · Method Simplify $\dfrac{2^3 \times 2^{-5}}{2^{-1}}$.
3. Core + · Method For $x \neq 0$, write $(x^2)^{-3} \times x^7$ in the form x^n.
4. Review · Quick Median of the series 4, 7, 9, 10, 12, 15?
5. Thread · Identify Which solid has a disc-shaped base, a curved lateral surface and a single vertex?

Sheet 32 Mon 18 Jan. 2027 *Square roots; $\sqrt{a^2} = |a|$*

1. Warm-up · Quick Calculate $\sqrt{36}$ and $\sqrt{0.09}$.
2. Core · Method Write $\sqrt{72}$ in the form $a\sqrt{2}$ and calculate $\sqrt{(-5)^2}$.
3. Core + · True/false True or false? Justify. $\sqrt{9+16} = \sqrt{9} + \sqrt{16}$.
4. Review · Quick Calculate $(3^2)^3$.
5. Thread · True/false True or false? Justify. The diagonals of a rhombus are perpendicular.

Sheet 33 Tue 19 Jan. 2027 *Fractional expressions; expanded, reduced or factorised forms*

1. Warm-up · Quick Calculate $\dfrac{1}{2} + \dfrac{1}{3}$.
2. Core · Method Write as a single fraction $\dfrac{1}{x} + \dfrac{1}{x+1}$ (for $x \neq 0$ and $x \neq -1$).
3. Core + · Method Factorise $x^2 - 4$ then simplify $\dfrac{x^2-4}{x-2}$ for $x \neq 2$.
4. Review · Quick Calculate $2a - 3b$ for $a = 2$ and $b = -1$.
5. Thread · Quick $ABCDEFGH$ is a cuboid with $AB = 4$, $AD = 3$, $AE = 2$. In the frame (A, B, D, E), what are the coordinates of G?

Sheet 34 Thu 21 Jan. 2027 *Inequalities: sum, multiplication by a real number*

1. Warm-up · Quick Complete with $<$ or $>$: $-3 \ \ldots \ -2$; $\dfrac{1}{3} \ \ldots \ 0.3$.
2. Core · Method We have $a < b$. Compare $-2a$ and $-2b$, then $a + 5$ and $b + 5$.
3. Core + · Method Given $2 < x < 3$ and $1 < y < 4$, find bounds for $x + y$ then $x - y$.
4. Review · Quick Calculate $(-3) \times (-4) + 2 \times (-5)$.
5. Thread · Problem A 120 cm by 84 cm sheet is cut into identical squares, with no waste. What is the largest possible side length?

Sheet 35 Mon 25 Jan. 2027 *Comparing by difference and by quotient*

1. Warm-up · Quick Compare $\dfrac{2}{3}$ and $\dfrac{3}{4}$ by studying the sign of the difference.
2. Core · Method Compare $x^2 + 1$ and $2x$ for every real number x.
3. Core + · Method Compare $\dfrac{7}{5}$ and $\dfrac{8}{6}$ by quotient (positive numbers).
4. Review · Table Monthly rainfall (fictional data).

Month	Jan.	Feb.	Mar.	Apr.
Rain (mm)	280	300	240	120

Which month is the wettest? What is the total over these 4 months?

5. Thread · Table → formula x: -2, -1, 0, 1 and $f(x)$: 4, 1, 0, 1. Is the function affine? Suggest a formula.

Sheet 36 Tue 26 Jan. 2027 *First-degree equations and inequalities; $x^2 = a$; zero product*

1. Warm-up · Quick Solve $3x - 5 = 7$.
2. Core · Method Solve $2x + 3 \leqslant 7 - x$.
3. Core + · Method Solve $(x - 2)(2x + 6) = 0$ then $x^2 = 7$.
4. Review · Quick Compare 1.4 and $\sqrt{2}$ without a calculator.
5. Thread · Quick ABC is isosceles at A with $\widehat{BAC} = 40°$. What is the measure of $\widehat{ABC}$?

Sheet 37 Thu 28 Jan. 2027 *Quotient equation $\frac{A(x)}{B(x)} = k$*

1. Warm-up · Quick For which value of x is the expression $\dfrac{1}{x-3}$ undefined?
2. Core · Method Solve $\dfrac{2x-1}{x+3} = 0$.
3. Core + · Method Solve $\dfrac{x+1}{x-2} = 3$.
4. Review · Expand Expand and simplify $(x + 2)(x - 5)$.
5. Thread · Code reading What do $g(3)$ and $g(-5)$ return?

```
def g(x):
    if x > 0:
        return 2*x
    else:
        return -x
```

Sheet 38 Mon 1st Feb. 2027 *Sign of a product or quotient; sign tables*

1. Warm-up · Quick Sign of $2x - 4$?
2. Core · Method Draw the sign table of $(x - 1)(x + 3)$.
3. Core + · Method Solve $\dfrac{x+2}{x-1} \geqslant 0$.
4. Review · Quick Pack of 32 cards: probability of drawing a heart? a king?
5. Thread · Disprove Disprove: "for every real number x, $x^2 \geqslant x$".